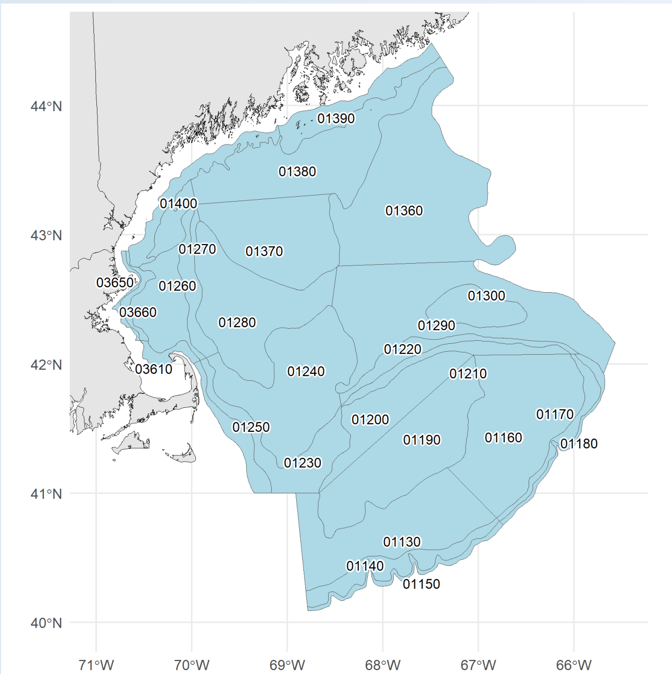
American Plaice (*Hippoglossoides platessoides*) Snapshot Ecosystem & Socioeconomic Profile

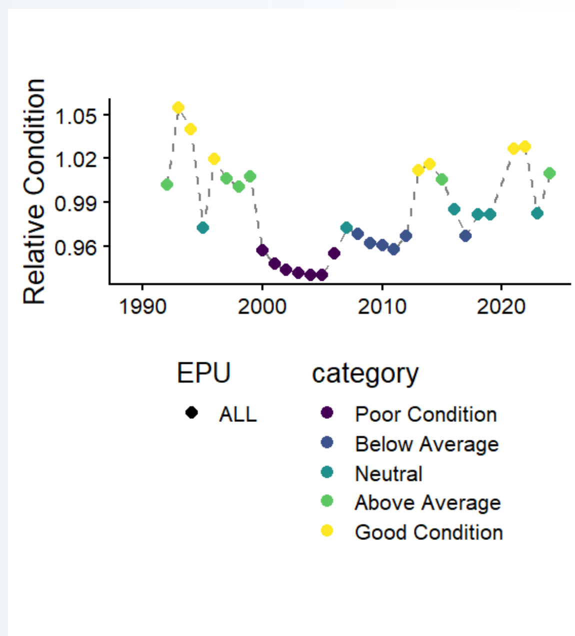
Fall 2026



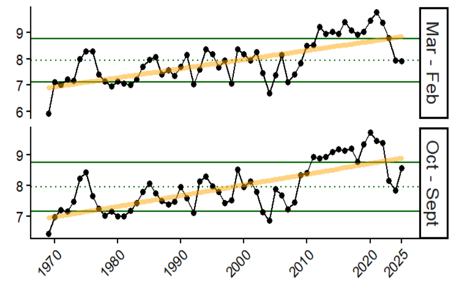
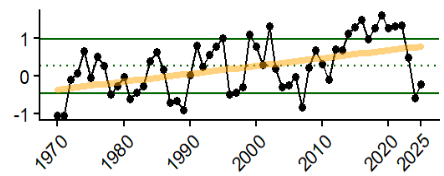
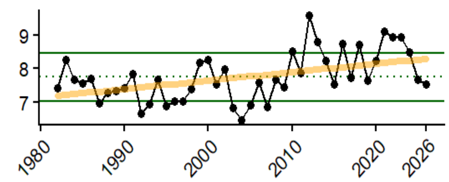
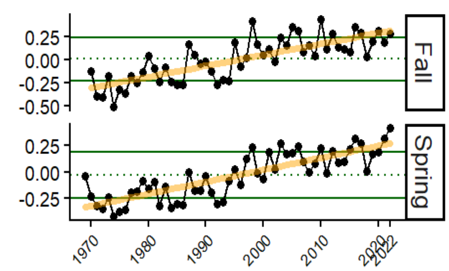
This snapshot ESP is provides an update to ecosystem indicators identified in the 2022 American Plaice Research Track Assessment Behan et al. (2022). This previous work found that the recruitment of plaice is related to sea surface temperature, the Atlantic Multidecadal Oscillation (AMO), and the North Atlantic Oscillation (NAO). AMO was also correlated with plaice condition. Bottom temperature anomalies, the Gulf Stream Index, and spawning stock biomass (SSB) were associated with plaice distribution. The goal of this report is to update the indicator datasets and provide context around what recent indicator data may mean for the American plaice stock, as well as highlighting recent socioeconomic information.

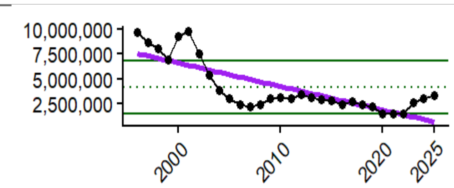
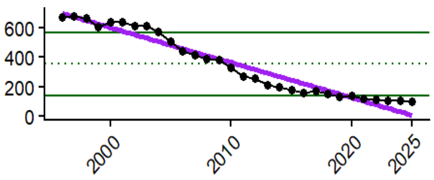
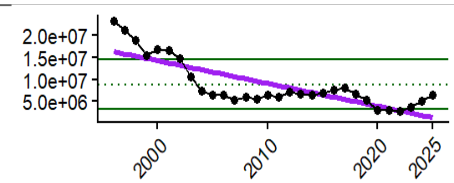
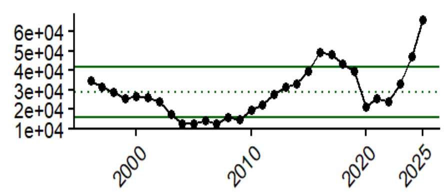
In the 2016 Climate Vulnerability Assessment, American plaice had an overall low vulnerability rank but high climate exposure from increasing surface temperatures and ocean acidification (Hare et al. 2016). There is evidence of American plaice shifting distribution into deeper waters (Nye et al. 2009), and further warming may decrease available habitat.

The New England Fishery Management Council (NEFMC) put American plaice under a rebuilding plan in 2004 (New England Fishery Management Council 2003) after significant declines in landings and revenue. While the number of commercial vessels landing plaice has continued to decline, revenue and landings have rebounded somewhat post-COVID19 (New England Fishery Management Council 2026) along with spawning stock biomass (Northeast Fisheries Science Center 2024).



Condition of American plaice across the Northeast Shelf exhibited lows in the 2000s to 2010, indicating thinner, lower-quality fish. Condition values have increased overall since 2010, with most recent years having neutral to good condition, and are experiencing fewer years of poor or below-average condition compared to lows in the 2000s.

Indicator Units	Status In 2026	Implications	Time Series
Bottom temperature (°C)	Near the long term mean (2025)	Bottom temperature impacts spawning, fish condition, and distribution. Optimum spawning temperatures range between 3-6°C but can tolerate -0.5 to 13°C. Annual bottom temperature was assessed over the year leading up to the spring and fall bottom trawl survey seasons. Recent annual mean bottom temperature is near the long-term mean. However, an overall warming trend in bottom temperatures may eventually result in changes in plaice distribution, spawning, and condition.	
Gulf Stream Index	Near the long term mean (2025)	The Gulf Stream affects plaice recruitment. The Gulf Stream Index was assessed over the year preceeding the fall bottont trawl surcey. There is an overall long-term increasing trend in the position of the Gulf Stream, with the Gulf Stream trending further north, which may be contributing to warmer water temperatures in the Gulf of Maine and affecting plaice recruitment.	
Sea Surface Temperature (Mar-June, °C)	Near the long term mean (2026)	March-June sea surface temperature affects plaice recruitment, as the thermal limit for survival and incubation of American plaice eggs is 14°C; sea surface temperatures have not reached this threshold value. Although recent cooler temperatures may be beneficial for plaice recruitment, the long-term increasing trend in sea surface temperatures may eventually result in negative impacts on recruitment.	
Atlantic Multidecadal Oscillation (AMO)	Above the long term mean (2022)	AMO values below 0 and above 0.25 are associated with lower recruitment, while values between approximately 0-0.25 are associated with higher recruitment (Behan and Kerr 2022, fig. 2). The AMO was assessed over the six months preceeding the spring and fall bottom trawl surveys. Recent AMO values are positive, with some being above the transition point of ~0.25, so impacts on recruitment may be mixed. Data since 2022 is not yet available.	

Indicator Units	Status In 2026	Implications	Time Series
Commerical landings (millions of lbs)	Near the long term average	Commercial landings of American plaice exhibited a sharp decline in the early 2000s with a long-term decreasing trend (purple). Trends in landings follow the implementation of a rebuilding plan for the stock in 2004 and successful rebuild in 2019.	
Number of commercial vessels (# of federally-permitted vessels landing over 1lb of American plaice)	Below the long term average	The number of commercial vessels landing American plaice has declined steadily from highs in the late 1990s, with a long-term decreasing trend (purple).	
Commercial revenue (millions, inflation adjusted to 2025 USD)	Near the long term average	Commercial revenue declined sharply after 2000 with a long-term decreasing trend (purple), likely due to a similar trend in landings. Average price per pound (not shown) dropped 50% from 2016 to 2023 and may also be a driver of declining revenue.	
Commercial revenue per active American plaice vessel (2025 USD)	Above the long term average	Revenue per vessel gradually increases from lows near 2010 to highs above the long-term mean prior to 2020. A steep decline in revenue per vessel in 2020 likely stems from the COVID-19 pandemic. In recent years, revenue per vessel has risen greatly to above the long-term mean, a possible result of a continued decline in vessels but slight increase in landings.	

* The y-axis units are included in the "Indicator" column of the table. In all figures, the dashed line represents the time series mean, and the solid green lines indicate ± 1 standard deviation. Commercial data were derived from the commercial dealer database hosted at the Greater Atlantic Regional Office. All dollar values have been adjusted to 2026 real dollars using the Gross Domestic Implicit Price Deflator.

We welcome your observations! Please contact northeast.ecosystem.highlights@noaa.gov with any on-the-water insights or changes observed in the American plaice fishery and nefsc.esp.leads@noaa.gov with questions or comments on the information presented in this report.

The code used to create this report can be viewed online: github.com/NEFSC/READ-EDAB-plaiceESP2026

References

Behan, Jamie, and Lisa Kerr. 2022. "Environmental Influences on American Plaice Stock Dynamics." Non-series Report. NOAA National Marine Fisheries Service, Northeast Fisheries Science Center. https://doi.org/10.25923/yvhh-4288.

Behan, Jamie, Lisa Kerr, Amanda Hart, Alex Hansell, Tyler Paklovitch, and Steve Cadrin. 2022. "Ecosystem Profile of American Plaice." Non-series Report. NOAA National Marine Fisheries Service, Northeast Fisheries Science Center. https://doi.org/10.25923/7kj5-1155.

Hare, Jonathan A., Wendy E. Morrison, Mark W. Nelson, Megan M. Stachura, Eric J. Teeters, Roger B. Griffis, Michael A. Alexander, et al. 2016. "A Vulnerability Assessment of Fish and Invertebrates to Climate Change on the Northeast U.S. Continental Shelf." *PLoS ONE* 11 (2): e0146756. https://doi.org/10.1371/journal.pone.0146756.

New England Fishery Management Council. 2003. "Final Amendment 13 to the Northeast Multispecies Fishery Management Plan Including a Final Supplemental Environmental Impact Statement (SEIS)." Vol. I–II. Newburyport, MA: New England Fishery Management Council. https://d23h0vhsm26o6d.cloudfront.net/Final-Amendment-13-SEISVol.-I-II.pdf.

———. 2026. "Amendment 25 (Revised) to the Northeast Multispecies Fishery Management Plan: Including an Environmental Assessment, Regulatory Flexibility Analysis, and Stock Assessment and Fishery Evaluation." Newburyport, MA: New England Fishery Management Council. https://d23h0vhsm26o6d.cloudfront.net/260305_Groundfish_Revised-A25_Final-Submission.pdf.

Northeast Fisheries Science Center. 2024. "American Plaice 2024 Management Track Assessment Report." Draft Working Paper. Woods Hole, MA: National Oceanic; Atmospheric Administration, National Marine Fisheries Service, Northeast Fisheries Science Center. https://apps-st.fisheries.noaa.gov/sis/docServlet?fileAction=download&fileId=9879.

Nye, Janet A., Jason S. Link, Jonathan A. Hare, and William J. Overholtz. 2009. "Changing Spatial Distribution of Fish Stocks in Relation to Climate and Population Size on the Northeast United States Continental Shelf." *Marine Ecology Progress Series* 393 (October): 111–29. https://doi.org/10.3354/meps08220.